

1) Scenario 1: Student Academic Performance Dashboard

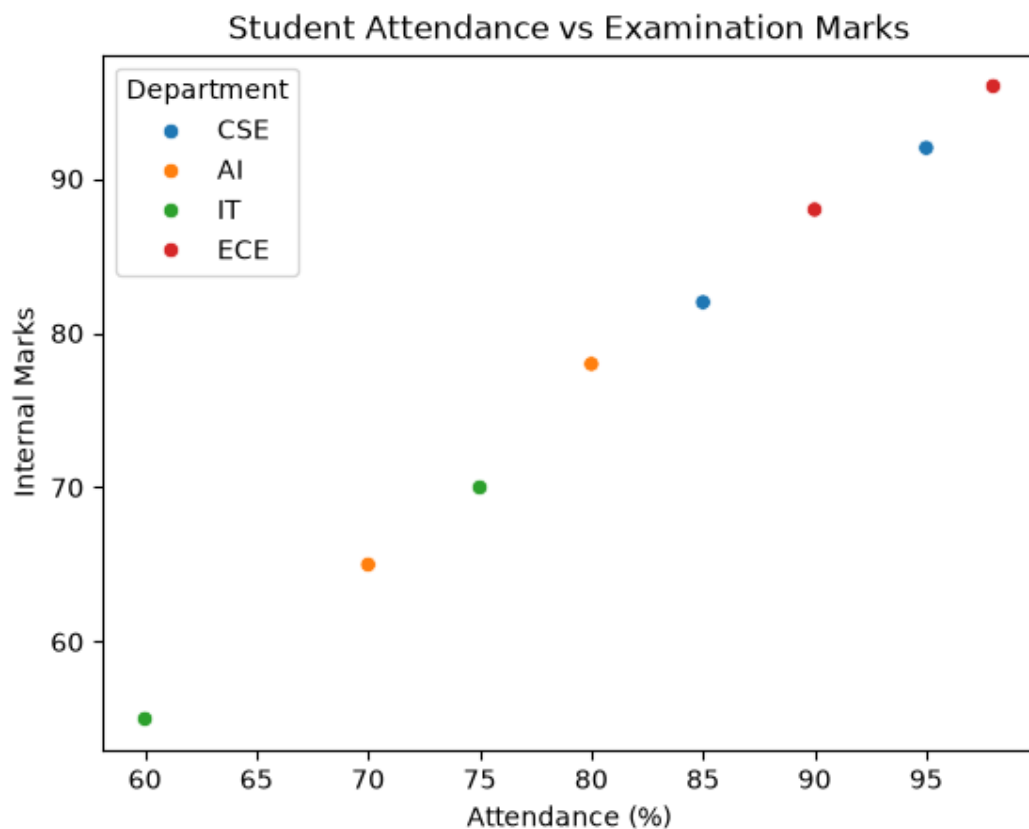
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Student ID":[1,2,3,4,5,6,7,8],
    "Attendance":[95,80,75,90,60,85,70,98],
    "Marks":[92,78,70,88,55,82,65,96],
    "Department":["CSE","AI","IT","ECE","IT","CSE","AI","ECE"]
}

df=pd.DataFrame(data)

sns.scatterplot(
    data=df,
    x="Attendance",
    y="Marks",
    hue="Department"
)

plt.title("Student Attendance vs Examination Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend(title="Department")
plt.show()
```



2) Scenario 2: Electric Vehicle Market Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Electric Vehicle Dataset
data = {
    "Vehicle Model": ["Model A", "Model B", "Model C", "Model D", "Model E", "Model F"],
    "Battery Capacity": [40, 60, 75, 90, 110, 130],
    "Price": [25000, 35000, 45000, 55000, 70000, 85000],
    "Driving Range": [250, 350, 450, 520, 650, 750],
    "Manufacturer": ["Tesla", "Tata", "Hyundai", "Tesla", "BMW", "Tata"]
}
```

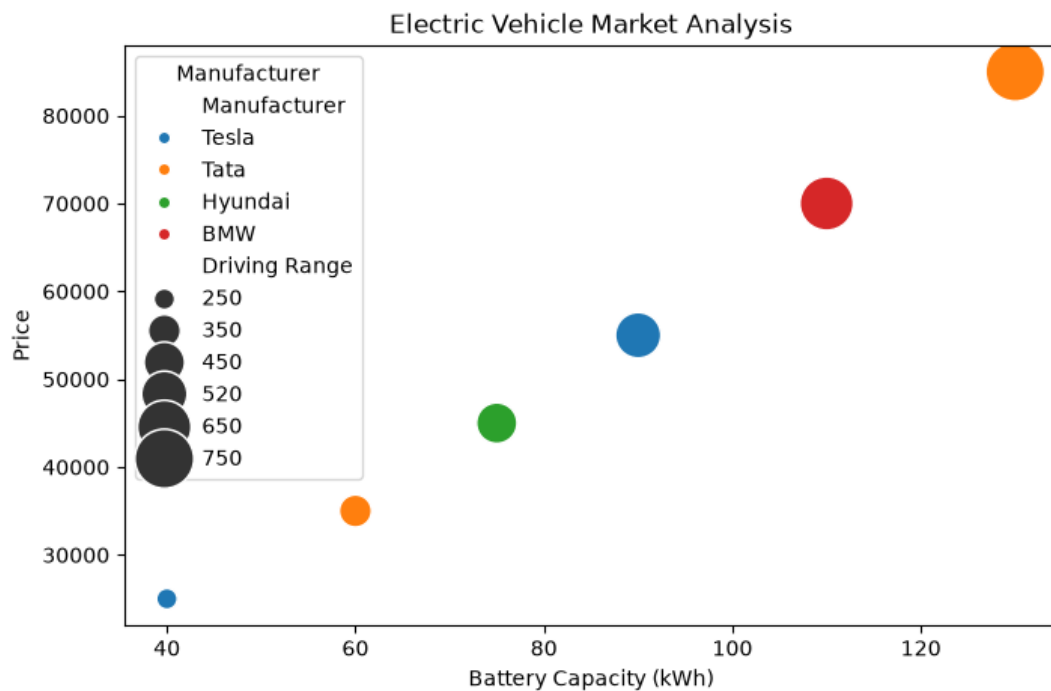
```
df = pd.DataFrame(data)
```

```
# Bubble Scatter Plot
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x="Battery Capacity",
    y="Price",
    size="Driving Range",
    hue="Manufacturer",
    sizes=(100,800)
)

plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price")
plt.legend(title="Manufacturer")

plt.show()
```



3) hospital patient admission analysis

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

Hospital Patient Admission Dataset

```
data = {
    "Department": ["Cardiology", "Neurology", "Orthopedics", "Pediatrics", "General"],
    "Number of Patients": [120, 90, 150, 80, 200],
    "Average Treatment Cost": [50000, 45000, 30000, 25000, 15000]
}
```

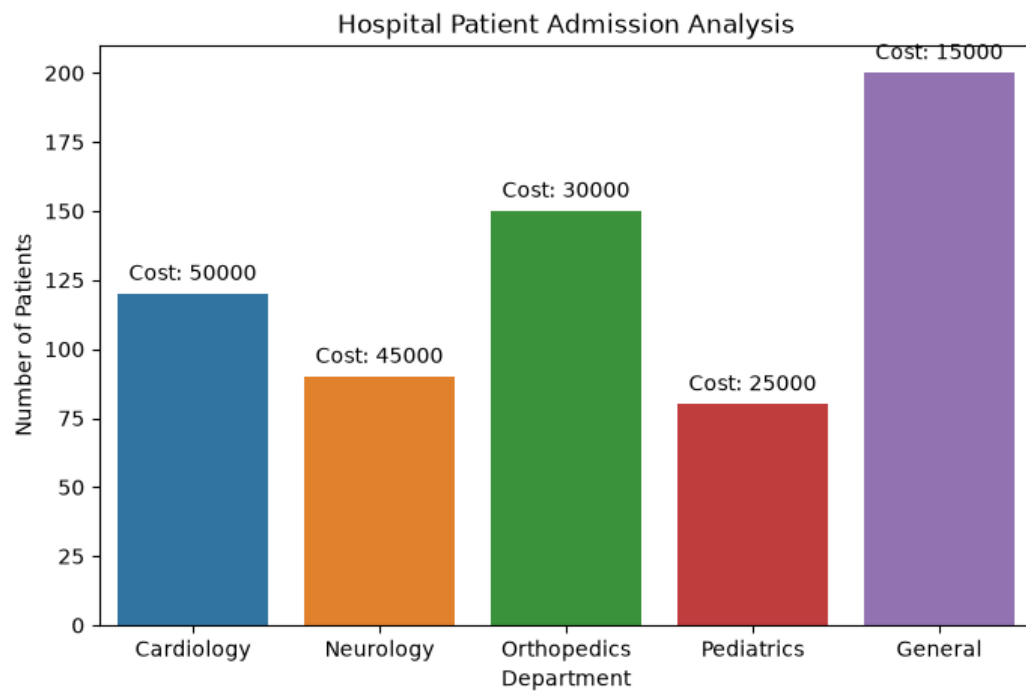
```
df = pd.DataFrame(data)
```

Bar Chart

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(
```

```
data=df,  
x="Department",  
y="Number of Patients",  
hue="Department"  
)  
  
# Add treatment cost labels  
for i, cost in enumerate(df["Average Treatment Cost"]):  
    plt.text(  
        i,  
        df["Number of Patients"][i] + 5,  
        "Cost: " + str(cost),  
        ha="center"  
    )  
  
plt.title("Hospital Patient Admission Analysis")  
plt.xlabel("Department")  
plt.ylabel("Number of Patients")  
  
plt.show()
```



4) Scenario 4: Weather Monitoring System

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Weather Dataset
```

```
data = {  
    "Date": [  
        "Day 1","Day 2","Day 3","Day 4","Day 5",  
        "Day 6","Day 7","Day 8","Day 9","Day 10"  
    ],  
    "Temperature": [  
        30,32,31,33,34,35,33,32,31,30  
    ],  
    "City": [  
        "Chennai","Chennai","Chennai","Chennai","Chennai",  
        "Delhi","Delhi","Delhi","Delhi","Delhi"  
    ]  
}
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Add another city data
```

```
data2 = {  
    "Date": [  
        "Day 1","Day 2","Day 3","Day 4","Day 5",  
        "Day 6","Day 7","Day 8","Day 9","Day 10"  
    ],  
    "Temperature": [  
        25,26,28,27,29,30,31,29,28,27  
    ],  
    "City": [  
        "Mumbai","Mumbai","Mumbai","Mumbai","Mumbai",  
        "Mumbai","Mumbai","Mumbai","Mumbai","Mumbai"  
    ]  
}
```

```
df2 = pd.DataFrame(data2)
```

```
# Combine data
```

```
df = pd.concat([df, df2])
```

```
# Line Chart
```

```
plt.figure(figsize=(9,5))
```

```
sns.lineplot(  
    data=df,  
    x="Date",  
    y="Temperature",
```

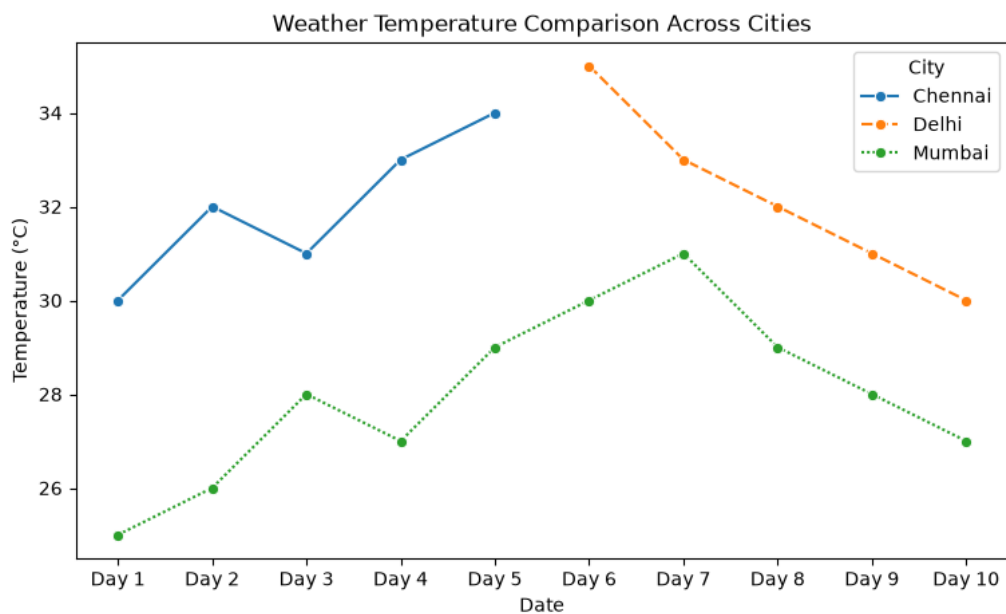
```

    hue="City",
    style="City",
    marker="o"
)

plt.title("Weather Temperature Comparison Across Cities")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")

plt.show()

```



5) Scenario 5: Supermarket Sales Analysis

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Supermarket Sales Dataset
data = {
    "Month": ["January","January","January","February","February","February",
              "March","March","March"],
    "Product Category": ["Electronics","Grocery","Clothing",

```



```
        "Electronics","Grocery","Clothing",
        "Electronics","Grocery","Clothing"],
    "Sales Amount": [50000,70000,45000,
                     60000,80000,50000,
                     75000,90000,65000],
    "Branch": ["Branch A","Branch A","Branch A",
               "Branch B","Branch B","Branch B",
               "Branch C","Branch C","Branch C"]
}
```

```
df = pd.DataFrame(data)
```

```
# Grouped Bar Chart
```

```
plt.figure(figsize=(9,5))
```

```
sns.barplot(
```

```
    data=df,
```

```
    x="Product Category",
```

```
    y="Sales Amount",
```

```
    hue="Branch"
```

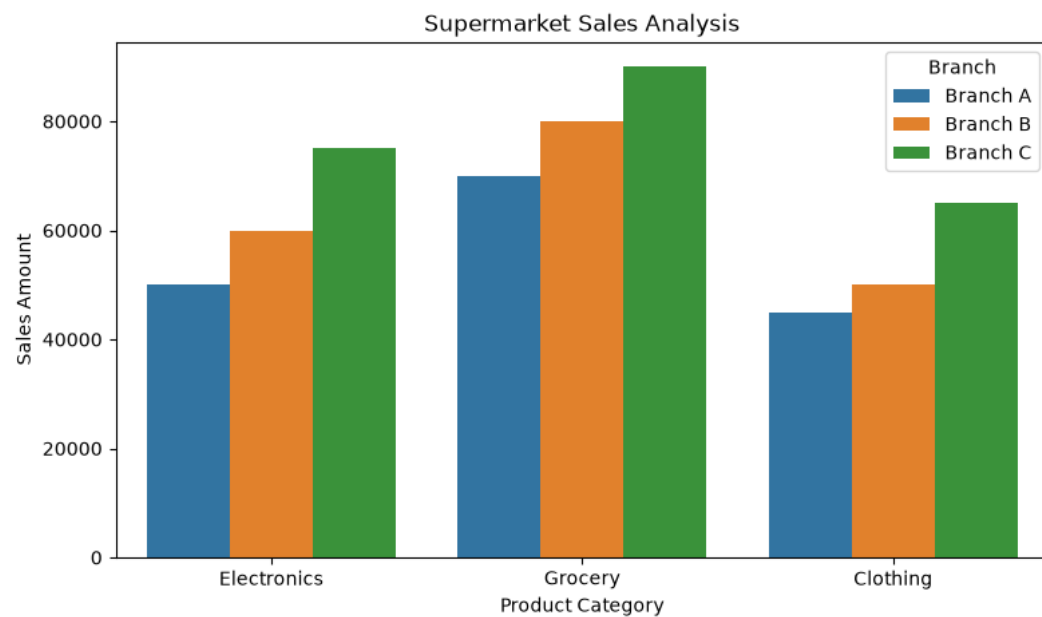
```
)
```

```
plt.title("Supermarket Sales Analysis")
```

```
plt.xlabel("Product Category")
```

```
plt.ylabel("Sales Amount")
```

```
plt.show()
```



6) Scenario 6: Employee Salary Distribution

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Employee Salary Dataset
```

```
data = {
    "Employee ID": [101,102,103,104,105,106,107,108,109,110,
                    111,112,113,114,115],
    "Salary": [25000,30000,35000,40000,45000,
               50000,55000,60000,65000,70000,
               75000,80000,85000,90000,100000],
    "Experience": [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15],
    "Department": ["IT","HR","IT","Finance","Sales",
                   "IT","HR","Finance","Sales","IT",
                   "HR","Finance","Sales","IT","Finance"]
}
```

```
df = pd.DataFrame(data)
```

```
# Histogram of Salaries
```

```
plt.figure(figsize=(8,5))
```

```
sns.histplot(
```

```
    data=df,
```

```
    x="Salary",
```

```
    bins=5
```

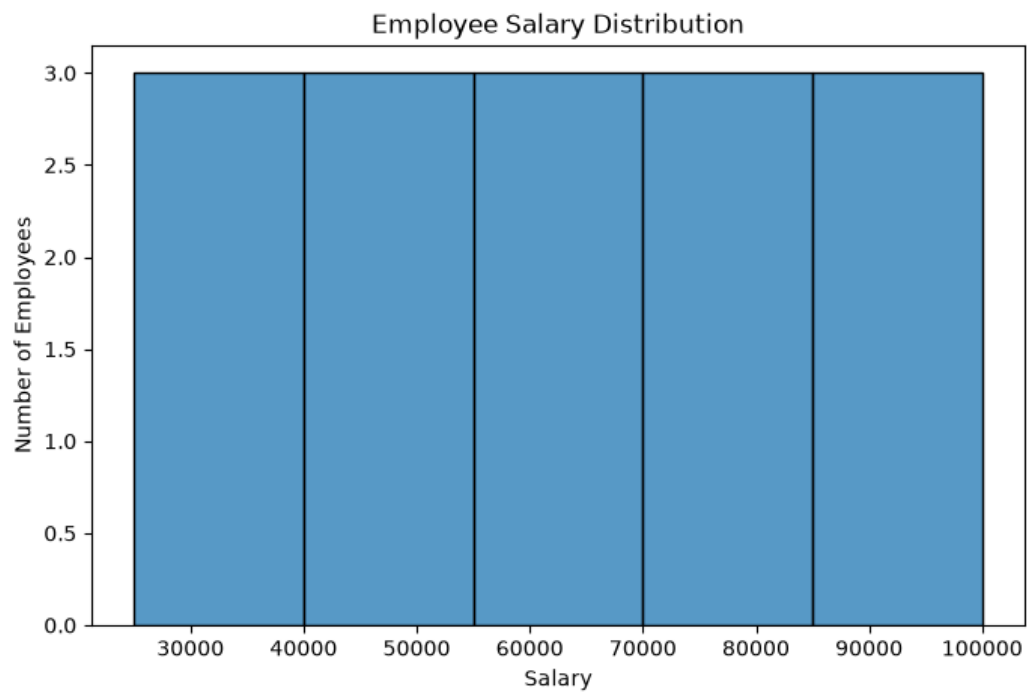
```
)
```

```
plt.title("Employee Salary Distribution")
```

```
plt.xlabel("Salary")
```

```
plt.ylabel("Number of Employees")
```

```
plt.show()
```



7) Scenario 7: Smartphone Market Comparison

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Brand": ["Apple","Samsung","OnePlus","Google","Xiaomi","Vivo"],
    "RAM": [6,8,12,8,12,8],
    "Processor Speed": [3.2,3.0,3.1,2.9,2.8,2.7],
    "Price": [90000,70000,50000,65000,30000,25000],
    "Storage": [128,256,256,128,512,256]
}

df = pd.DataFrame(data)

plt.figure(figsize=(8,5))

shape_map = {
    "Apple": "o",
    "Samsung": "s",
    "OnePlus": "^",
    "Google": "*",
    "Xiaomi": "X",
    "Vivo": "D"
}

sns.scatterplot(
    data=df,
    x="RAM",
    y="Price",
    hue="Processor Speed",
    style="Brand",
```

```

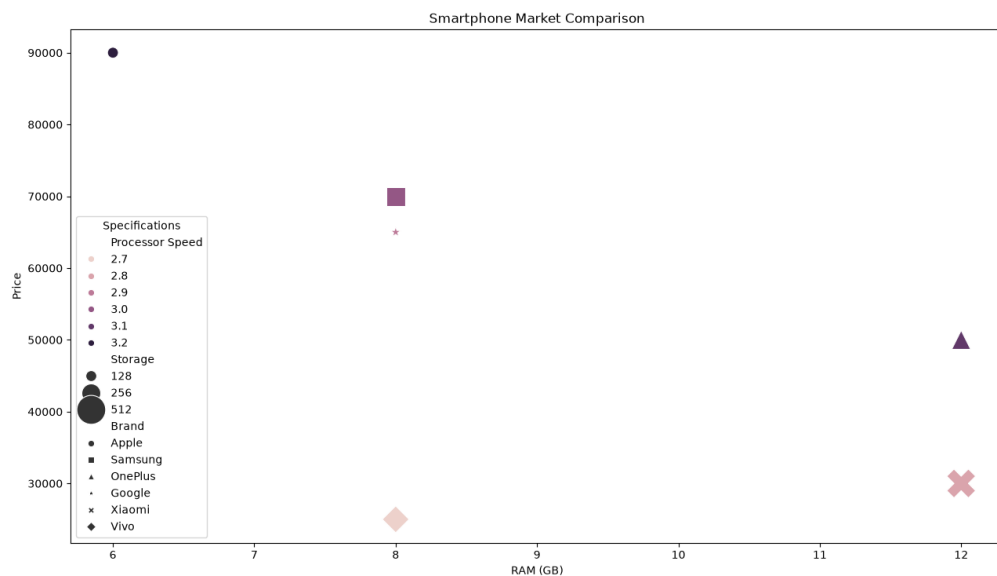
markers=shape_map,
size="Storage",
sizes=(100,700)
)

plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price")

plt.legend(title="Specifications")

plt.show()

```



8) Scenario 8: Online Movie Rating Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Movie Rating Dataset
data = {
    "Genre": ["Action", "Comedy", "Drama", "Horror", "Sci-Fi"],
    "Average Rating": [8.2, 7.5, 8.8, 6.9, 8.5],
    "Number of Reviews": [5000, 3500, 6000, 2500, 4500],
    "Release Year": [2024, 2024, 2024, 2024, 2024]
}

df = pd.DataFrame(data)

# Create heatmap data
heatmap_data = df.pivot_table(
    index="Genre",
    values="Average Rating"
)

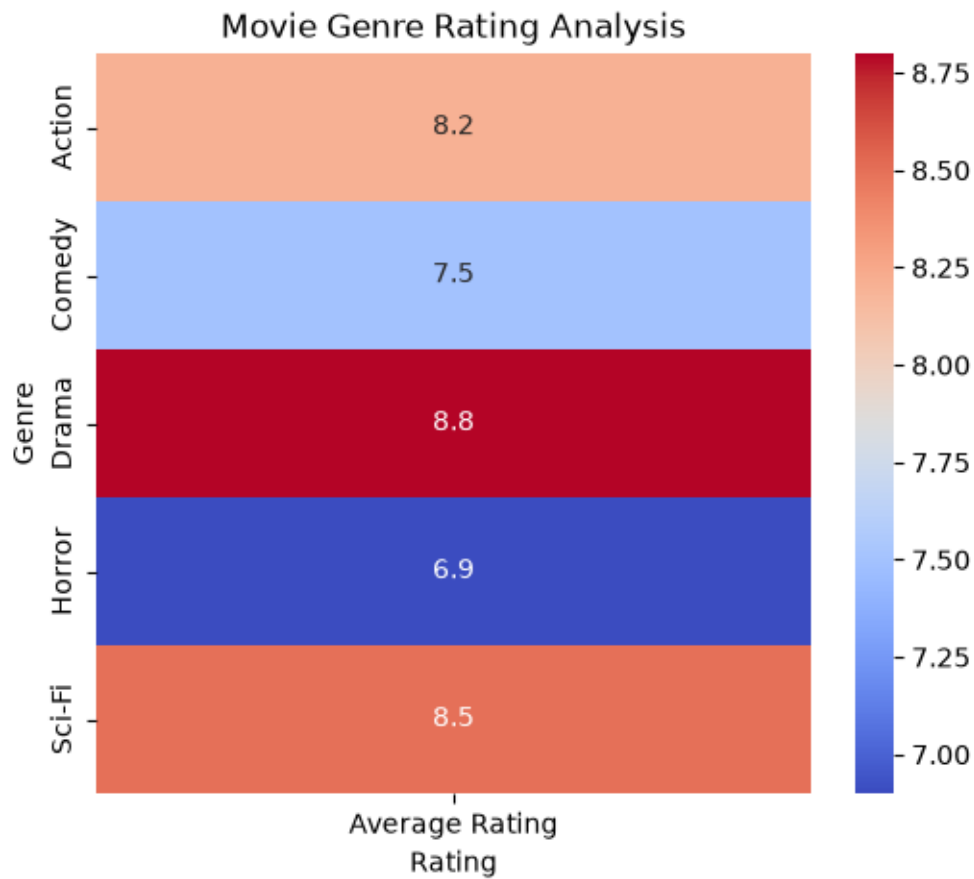
# Heatmap
plt.figure(figsize=(6,5))

sns.heatmap(
    heatmap_data,
    annot=True,
    cmap="coolwarm"
)

plt.title("Movie Genre Rating Analysis")
plt.xlabel("Rating")
```

```
plt.ylabel("Genre")
```

```
plt.show()
```



9) Scenario 9: COVID-19 State-wise Analysis

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# COVID-19 State-wise Dataset
```

```
data = {
```

```
    "State": ["Maharashtra", "Kerala", "Karnataka", "Tamil Nadu", "Delhi", "Gujarat"],
```

```
    "Active Cases": [25000, 18000, 15000, 12000, 9000, 7000],
```

```
    "Vaccination Percentage": [85, 90, 88, 92, 95, 87],
```

```
    "Population": [125, 35, 68, 78, 20, 70]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Tile Plot (Choropleth alternative)
```

```
plt.figure(figsize=(8,5))
```

```
sns.heatmap(
```

```
    df[["Active Cases"]].T,
```

```
    annot=True,
```

```
    cmap="Reds",
```

```
    xticklabels=df["State"],
```

```
    yticklabels=["Active Cases"]
```

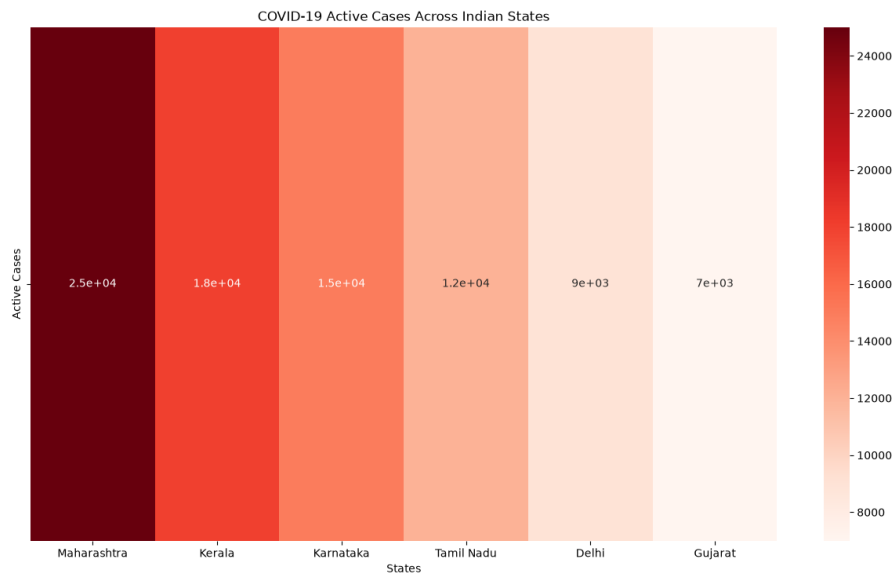
```
)
```

```
plt.title("COVID-19 Active Cases Across Indian States")
```

```
plt.xlabel("States")
```

```
plt.ylabel("")
```

```
plt.show()
```

10) Scenario 10: Iris Flower Classification

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load Iris Dataset
```

```
iris = sns.load_dataset("iris")
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(
```

```
    data=iris,
```

```
    x="sepal_length",
```

```
    y="petal_length",
```

```
    hue="species"
```

```
)
```

```
plt.title("Iris Flower Classification")
```

```
plt.xlabel("Sepal Length (cm)")
```

```
plt.ylabel("Petal Length (cm)")
```

```
plt.legend(title="Species")
```

```
plt.show()
```

